Internship Opportunity – Spatiotemporal Segmentation of DAS Vibration Events

Background

Distributed Acoustic Sensing (DAS) turns an optical fibre into a continuous array of acoustic sensors capable of measuring vibrations over several kilometres. Its operation relies on Rayleigh backscattering: when a light pulse travels through the fibre, a small fraction of the light is scattered back towards the source due to microscopic irregularities in the glass. The phase and amplitude of the backscattered light change when the fibre is strained, making it possible to detect and locate vibrational events.
The SURFO team uses this technology to monitor railway infrastructure. Applications include intrusion detection near railway tracks, identification of rockfalls and monitoring train movements. Segmentation of vibrational events is a common step for identifying regions of interest in DAS data and feeding the subsequent processing tasks specific to each application.

Challenge

Events have signatures that vary in shape, duration, spatial extent and intensity. Their identification is complicated by background noise, multiple simultaneous sources, and changes in measurement conditions and acquisition settings. The challenge is to segment these signatures while preserving their spatiotemporal context, despite limited annotations and possible class imbalance between applications.

Internship objective

Develop and evaluate signal-processing and/or artificial-intelligence methods to distinguish regions associated with vibrational events from background noise and delineate their spatial and temporal extent. The work aims to establish a common segmentation framework across applications. Depending on the available annotations, an extension to multiclass segmentation may also be investigated.

Internship tasks

The work will focus on several complementary topics:

- **Analyse existing data and methods**: become familiar with DAS data, characterise the signatures of target events and review the literature on spatiotemporal segmentation.
- **Prepare a reference dataset**: define annotation rules, create segmentation masks and select suitable representations (preprocessed signals, envelopes or energy maps).
- **Develop and compare approaches**: establish a signal- or image-processing baseline, then evaluate a learning-based method (U-Net, ResUNet, AnisotropicResUNet) adapted to the number of annotations available.
- **Assess robustness**: evaluate noise, overlapping events, spatial and temporal resolutions, and physical scales (metres and seconds); assess IoU, Dice, precision, recall, false alarms and computation time.
- **Consolidate the developments**: provide documented Python code, reproducible experiments and an assessment of performance and limitations to support integration into the team's tools.

The internship will initially focus on preparing the reference dataset, comparing methods and validating them. Multiclass segmentation and streaming processing may be explored further depending on progress and results.

Expected outcomes

- A documented reference dataset and evaluation protocol;
- A prototype producing segmentation masks and spatiotemporal boundaries of detected regions;
- A comparison of performance across applications and recommendations for future work.

Candidate profile

Education

Final-year engineering student, second-year Master's student or student on a gap year, with a background in signal processing, artificial intelligence or applied mathematics.

Technical skills

- Strong foundations in mathematics, signal processing and/or *machine learning*;
- Proficiency in Python (NumPy, SciPy and PyTorch are appreciated);
- Ability to analyse experimental data.

Personal qualities

- Scientific rigour, curiosity and independence;
- Teamwork and good communication skills.

Practical information

- Duration: 5 to 6 months
- Location: La Plaine-Saint-Denis, France
- Start date: February–March 2027
- Supervision: RIHANI Mahran, BEN AMARA Imen
- Allowance: in accordance with applicable regulations
- Benefits: SNCF travel benefits.

How to apply

Please send your CV and cover letter to all the following addresses:

- `ext.mahran.rihani@reseau.sncf.fr`
- `gabriel.papaiz@reseau.sncf.fr`
- `ext.imen.benamara@reseau.sncf.fr`

References

- Xie D. et al. (2025). *Intelligent Traffic Monitoring with Distributed Acoustic Sensing.* Seismological Research Letters, 96(4), 2477–2488. doi:10.1785/0220240298.
- Peng F., Zhu Z., Zhang Y. and Miao Q. (2025). *A Deep Learning Image Segmentation Model for Detection of Weak Vehicle-Generated Quasi-Static Strain in Distributed Acoustic Sensing.* IEEE Transactions on Intelligent Transportation Systems, 26(6), 8933–8944.
- Zhong Y. et al. (2025). *Acoustic Events Detection and Classification for Urban DAS Data with Supervised Learning.* Seismological Research Letters, 96(5), 3112–3131. doi:10.1785/0220240222.